CONFORMAL SIGNAL TEMPORAL LOGIC FOR ROBUST REINFORCEMENT LEARNING CONTROL

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Motivation

Reinforcement learning is powerful, but safety remains difficult under stress.

- RL can control nonlinear aerospace systems.
- Learned policies may fail under disturbances, actuator limits, delay, noise and model mismatch.
- Safety-critical systems require runtime guarantees beyond average performance metrics.

Case Study: AeroBench F–16 Engine Control

We study safe outer-loop control for the F–16 engine benchmark.

- State:

$$s_t = [V_t,\ pow_t]^\top$$

- Action:

$$a_t \in [0, 1]$$

- Objective: track commanded airspeed setpoint sp.
- Controller: PPO policy trained under nominal conditions.

Signal Temporal Logic Requirement

The safety requirement is bounded tracking error during the final part of the episode:

$$\varphi_{\text{track}} := \mathbf{G}_{[T-W,T]}\left(\left|\frac{V_t - sp}{sp}\right| \le \tau\right)$$

STL Robustness: Why It Matters

STL does not only classify safety; it quantifies the safety margin.

- $\rho > 0$: specification satisfied with margin.
- $\rho < 0$: specification violated.
- The shield evaluates worst-case robustness under conformal uncertainty.

$$\rho^{\varphi}_{\min}(t) = \min_{x \in \mathcal{C}(\hat{x}_{t:t+H})} \rho^{\varphi}(x, t)$$

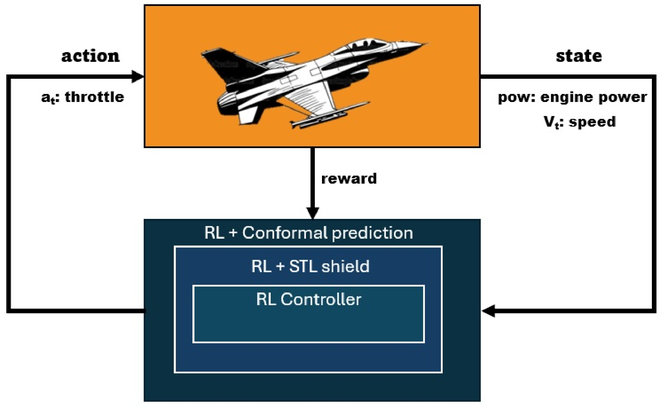
This enables the controller to quantify risk and proactively avoid unsafe actions before violations occur.

Proposed Framework

We integrate conformal prediction with STL monitoring to provide a runtime safety layer for reinforcement learning control.

A PPO policy generates throttle actions, while the conformal STL shield evaluates worst-case temporal-logic robustness and enforces safety under uncertainty.

Overall Control Loop



- The F–16 plant returns airspeed V_t and engine power pow_t.
- PPO proposes a throttle action.
- STL shielding checks temporal-logic safety.
- Conformal prediction adds uncertainty-aware robustness.

Conformal STL Shield

The one-step predictor estimates airspeed:

$$\widehat{V}_{t+1} = aV_t + b\,pow_t + c\,u_t + d$$

Split conformal prediction computes:

$$\hat{q}_{1-\delta} = \text{Quantile}_{1-\delta}\left(|V_{t+1} - \widehat{V}_{t+1}|\right)$$

This gives:

$$V_{t+1} \in [\widehat{V}_{t+1} - \hat{q}_{1-\delta}, \widehat{V}_{t+1} + \hat{q}_{1-\delta}]$$

The shield selects:

$$u_t \in \arg\max_{u \in \mathcal{U}(u_{t-1})} \rho^{\varphi_{\text{track}}}_{\min}(t; u)$$

$$\mathcal{U}(u_{t-1}) = \{u \in [0, 1] : |u - u_{t-1}| \le s\}$$

Stress Benchmark Evaluation

Trained at nominal conditions, the PPO policy is evaluated under a progressive stress benchmark without retraining.

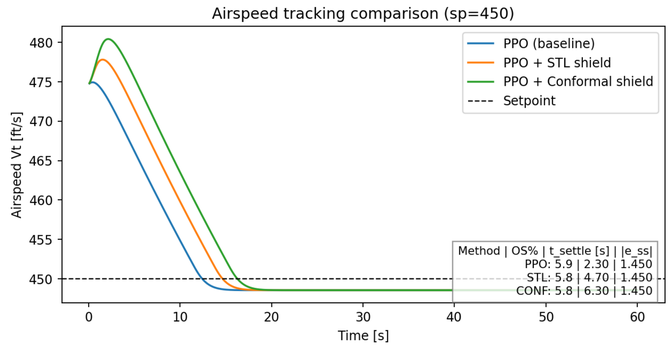
- **S0 Nominal:** no disturbance, 5% band, 10s window.
- **S1 Mild:** small rate limit and observation noise.
- **S2 Moderate:** throttle saturation, delay/noise.
- **S3 Strong:** model mismatch, throttle cap, noise, delay, and setpoint jump.

Progressive Stress Benchmark

Scenario	PPO	PPO+STL	PPO+CONF
S0 Nominal	100%	100%	100%
S1 Mild	100%	100%	100%
S2 Moderate	90%	95%	**100%**
S3 Strong	60%	75%	**95%**

S3: Satisfaction improves from **60%** to **95%**; mean robustness improves from **0.006** to **0.017**.

Airspeed Tracking



Discussion and Takeaway

- Under nominal conditions, all methods satisfy the STL requirement.
- Under strong stress, unshielded PPO drops to 60% satisfaction.
- The conformal STL shield shifts robustness toward positive values.
- Safety improves without retraining the RL policy.
- Runtime overhead was negligible in this benchmark.

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